

Chapter 3

Knowledge Representation and Reasoning

What is Knowledge Representation?

Humans are best at understanding, reasoning, and interpreting knowledge. Human knows things, which is knowledge and as per their knowledge they perform various actions in the real world. But, how machines do all these things comes under knowledge representation and reasoning. Hence, we can describe Knowledge representation as following:

Knowledge representation and reasoning (KR, KRR) is the part of Artificial intelligence which concerned with AI agents thinking and how thinking contributes to intelligent behavior of agents. It is responsible for representing information about the real world so that a computer can understand and can utilize this knowledge to solve the complex real world problems such as diagnosis a medical condition or communicating with humans in natural language.

It is also a way which describes how we can represent knowledge in artificial intelligence. Knowledge representation is not just storing data into some database, but it also enables an intelligent machine to learn from that knowledge and experiences so that it can behave intelligently like a human.

A knowledge-based agent

A knowledge-based agent includes a knowledge base and an inference system.

A knowledge base is a set of representations of facts of the world. Each individual representation is called a sentence. The sentences are expressed in a knowledge representation language.

The agent operates as follows:

1. It TELLS the knowledge base what it perceives.
 2. It ASKS the knowledge base what action it should perform.
 3. It performs the chosen action.
- Intelligent agents need knowledge about the world for making good decisions.
 - The knowledge of an agent is stored in a knowledge base in the form of **sentences** in a knowledge representation language.
 - A knowledge-based agent needs a **knowledge base** and an **inference mechanism**. It operates by storing sentences in its knowledge base, inferring new sentences with the inference mechanism, and using them to deduce which actions to take.
 - A **representation language** is defined by its syntax and semantics, which specify the structure of sentences and how they relate to the facts of the world.
 - The **interpretation** of a sentence is the fact to which it refers. If this fact is part of the actual world, then the sentence is true.

What to Represent:

Following are the kind of knowledge which needs to be represented in AI systems:

Object: All the facts about objects in our world domain. E.g., Guitars contains strings, trumpets are brass instruments.

Events: Events are the actions which occur in our world.

Performance: It describes behavior which involves knowledge about how to do things.

Meta-knowledge: It is knowledge about what we know.

Facts: Facts are the truths about the real world and what we represent.

Knowledge-Base: The central component of the knowledge-based agents is the knowledge base. It is represented as KB. The Knowledgebase is a group of the Sentences (Here, sentences are used as a technical term and not identical with the English language).

Knowledge: Knowledge is awareness or familiarity gained by experiences of facts, data, and situations.

Types of knowledge

The following are the various types of knowledge: Hence, Knowledge Representation in Artificial intelligence.....

1. **Declarative Knowledge:** Declarative knowledge is to know about something.

It includes concepts, facts, and objects. It is also called descriptive knowledge and expressed in declarative sentences. It is simpler than procedural language.

2. **Procedural Knowledge:** It is also known as imperative knowledge.

Procedural knowledge is a type of knowledge which is responsible for knowing how to do something. It can be directly applied to any task. It includes rules, strategies, procedures, agendas, etc. Procedural knowledge depends on the task on which it can be applied.

3. **Meta-knowledge:** Knowledge about the other types of knowledge is called Meta-knowledge.

4. **Heuristic knowledge:** Heuristic knowledge is representing knowledge of some experts in a field or subject. Heuristic knowledge is rules of thumb based on previous experiences, awareness of approaches, and which are good to work but not guaranteed.

5. **Structural knowledge:** Structural knowledge is basic knowledge to problem-solving. It describes relationships between various concepts such as kind of, part of, and grouping of something. It describes the relationship that exists between concepts or objects.

The relation between knowledge and intelligence:

Knowledge of real-worlds plays a vital role in intelligence and same for creating artificial intelligence. Knowledge plays an important role in demonstrating intelligent behavior in AI agents. An agent is only able to accurately act on some input when he has some knowledge or experience about that input.

Techniques of Knowledge Representation

There are mainly four ways of knowledge representation which are given as follows:

1. Logical Representation
2. Semantic Network Representation
3. Production Rules
4. Frame Representation

1. Logical Representation

Logical representation is a language with some concrete rules which deals with propositions and has no ambiguity in representation. Logical representation means drawing a conclusion based on various conditions. This representation lays down some important communication rules. It consists of precisely defined syntax and semantics which supports the sound inference. Each sentence can be translated into logics using syntax and semantics.

Syntax:

- Syntaxes are the rules which decide how we can construct legal sentences in the logic.
- It determines which symbol we can use in knowledge representation.
- How to write those symbols.

Semantics:

- Semantics are the rules by which we can interpret the sentence in the logic.
- Semantic also involves assigning a meaning to each sentence.

Logical representation can be categorized into mainly two logics:

- a. Propositional Logics
- b. Predicate logics

Advantages of logical representation:

- A. Logical representation enables us to do logical reasoning.
- B. Logical representation is the basis for the programming languages.

Disadvantages of logical Representation:

- A. Logical representations have some restrictions and are challenging to work with.
- B. Logical representation technique may not be very natural, and inference may not be so efficient.

Note: Do not be confused with logical representation and logical reasoning as logical representation is a representation language and reasoning is a process of thinking logically.

2. Semantic Network Representation

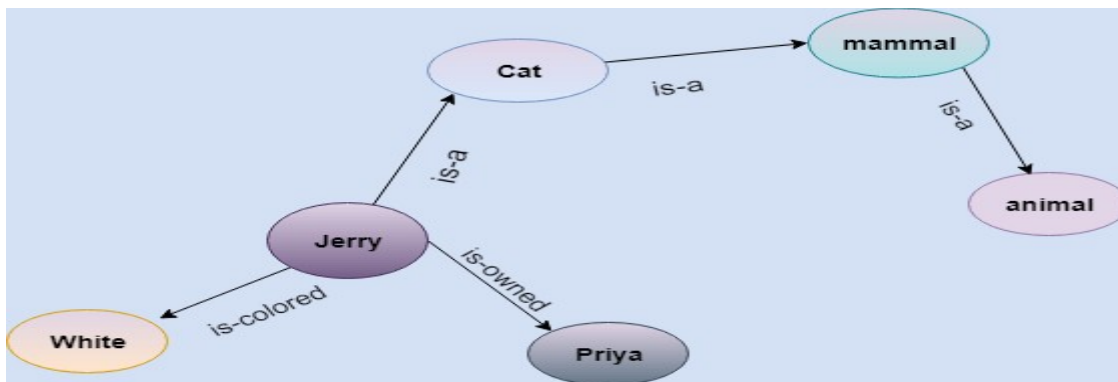
Semantic networks are alternative of predicate logic for knowledge representation. In Semantic networks, we can represent our knowledge in the form of graphical networks. This network consists of nodes representing objects and arcs which describe the relationship between those objects. Semantic networks can categorize the object in different forms and can also link those objects. Semantic networks are easy to understand and can be easily extended. This representation consists of mainly two types of relations:

- a. IS-A relation (Inheritance)
- b. Kind-of-relation

Example: Following are some statements which we need to represent in the form of nodes and arcs.

Statements:

- a. Jerry is a cat.
- b. Jerry is a mammal
- c. Jerry is owned by Priya.
- d. Jerry is white colored.
- e. All Mammals are animal.



In the above diagram, we have represented the different type of knowledge in the form of nodes and arcs. Each object is connected with another object by some relation.

Advantages of Semantic network:

- A. Semantic networks are a natural representation of knowledge.
- B. Semantic networks convey meaning in a transparent manner.
- C. These networks are simple and easily understandable.

Drawbacks in Semantic representation:

- A. Semantic networks take more computational time at runtime as we need to traverse the complete network tree to answer some questions. It might be possible in the worst case scenario that after traversing the entire tree, we find that the solution does not exist in this network.
- B. Semantic networks try to model human-like memory (Which has 10¹⁵ neurons and links) to store the information, but in practice, it is not possible to build such a vast semantic network.
- C. These types of representations are inadequate as they do not have any equivalent quantifier, e.g., for all, for some, none, etc.
- D. Semantic networks do not have any standard definition for the link names.
- E. These networks are not intelligent and depend on the creator of the system.

3. Production Rules

Production rules system consist of (condition, action) pairs which mean, "If condition then action". It has mainly three parts:

- The set of production rules
- Working Memory
- The recognize-act-cycle

In production rules agent checks for the condition and if the condition exists then production rule fires and corresponding action is carried out. The condition part of the rule determines which rule may be applied to a problem. And the action part carries out the associated problem-solving steps. This complete process is called a recognize-act cycle.

The working memory contains the description of the current state of problems-solving and rule can write knowledge to the working memory. This knowledge match and may fire other rules.

If there is a new situation (state) generates, then multiple production rules will be fired together, this is called conflict set. In this situation, the agent needs to select a rule from these sets, and it is called a conflict resolution.

Example:

IF (at bus stop AND bus arrives) THEN action (get into the bus)

IF (on the bus AND paid AND empty seat) THEN action (sit down).

IF (on bus AND unpaid) THEN action (pay charges).

IF (bus arrives at destination) THEN action (get down from the bus).

Advantages of Production rule:

- A. The production rules are expressed in natural language.

- B. The production rules are highly modular, so we can easily remove, add or modify an individual rule.

Disadvantages of Production rule:

- A. Production rule system does not exhibit any learning capabilities, as it does not store the result of the problem for the future uses.
- B. During the execution of the program, many rules may be active hence rule-based production systems are inefficient.

4. Frame Representation

A frame is a record like structure which consists of a collection of attributes and its values to describe an entity in the world. Frames are the AI data structure which divides knowledge into substructures by representing stereotypes situations. It consists of a collection of slots and slot values. These slots may be of any type and sizes. Slots have names and values which are called facets.

Facets: The various aspects of a slot are known as **Facets**. Facets are features of frames which enable us to put constraints on the frames. Example: IF-NEEDED facts are called when data of any particular slot is needed. A frame may consist of any number of slots, and a slot may include any number of facets and facets may have any number of values. A frame is also known as **slot-filter knowledge representation** in artificial intelligence.

Frames are derived from semantic networks and later evolved into our modern-day classes and objects. A single frame is not much useful. Frames system consists of a collection of frames which are connected. In the frame, knowledge about an object or event can be stored together in the knowledge base. The frame is a type of technology which is widely used in various applications including Natural language processing and machine visions.

Advantages of Frame Representation:

- A. The frame knowledge representation makes the programming easier by grouping the related data.
- B. The frame representation is comparably flexible and used by many applications in AI.
- C. It is very easy to add slots for new attribute and relations.
- D. It is easy to include default data and to search for missing values.
- E. Frame representation is easy to understand and visualize

Disadvantages of frame representation:

- A. In frame system inference mechanism is not be easily processed.
- B. Inference mechanism cannot be smoothly preceded by frame representation.
- C. Frame representation has a much generalized approach.